

General description

This dataset contains pot-level measurements from a fully factorial experiment manipulating watering intensity, watering frequency, and plant community composition to assess their effects on functional traits and fitness of *Bromus erectus*.

Each row represents one experimental pot. Trait values are aggregated at the pot level from multiple individuals. For detailed measurement protocols, trait definitions, and replication information, please refer to [Table S3](#) (functional traits) and [Table S4](#) (fitness traits) in the manuscript and Supporting Information.

Experimental factors

Variable	Description	Type	Levels
SampleID	Unique sample identifier	Character	
Pot	Pot number (experimental unit)	Character	
Intensity	Watering intensity (total water input)	Factor	Wet, Dry
Frequency	Watering frequency (events per month)	Factor	Low, High
Composition	Plant community composition	Factor	Conspecific, Heterospecific

Growth traits

Variable	Description	Units
N	Plant number per pot	count
H	Mean plant height	cm
SB	Shoot biomass per individual (per ramet)	g
RB	Root biomass per individual (per ramet)	g
TB	Total biomass per individual (per ramet)	g
R_S	Root-to-shoot biomass ratio	unitless

Leaf traits

Variable	Description	Units
SLA	Specific leaf area	cm ² g ⁻¹
LDMC	Leaf dry matter content	mg g ⁻¹
LCC	Leaf carbon content	%
LNC	Leaf nitrogen content	%
Leaf_CN	Leaf carbon-to-nitrogen ratio	unitless

Physiological traits

Variable	Description	Units
Chl	Chlorophyll content per leaf area	mg m ⁻²
PSIIeff	Efficiency of PSII photochemistry (Fv/Fm)	unitless
Ψ _{osm}	Leaf osmotic potential	MPa
SS	Stomatal size	μm
SD	Stomatal density	No. mm ⁻²

Root traits

Variable	Description	Units
RD	Mean root diameter	mm
RDMC	Root dry matter content	mg g ⁻¹
SRL	Specific root length	m g ⁻¹
SRA	Specific root area	cm ² g ⁻¹
RTD	Root tissue density	g cm ⁻³
RCC	Root carbon content	%
RNC	Root nitrogen content	%
Root_CN	Root carbon-to-nitrogen ratio	unitless

Fitness components

Variable	Description	Units
SER	Seedling establishment rate (2022)	proportion (0–1)
SSR1	Seedling survival rate (2022–2023)	proportion (0–1)
SSR2	Seedling survival rate (2023–2024)	proportion (0–1)
SP1	Seed production per sown seed (2023)	count per seed
SP2	Seed production per sown seed (2024)	count per seed
SP _{total}	Total seed production (2023–2024) per sown seed	count per seed
RP1	Ramet production per sown seed (2023)	count per seed
RP2	Ramet production per sown seed (2024)	count per seed
TBs	Total biomass (pot) per sown seed (2024)	g per seed

Biomass variable

Variable	Description	Units
TB _{pot}	Total community biomass per pot (covariate)	g